



GPL812P01A Programming Guide

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1 IMPORTANT NOTICE BEFORE WRITING YOUR PROGRAM

1.1 Confirmation Sheet

It is highly recommended programmers to obtain a copy of respective confirmation sheet before coding a program. The confirmation sheet, a requisite document before placing orders, contains useful information and checklist to help programmers avoiding mistakes during program development. Due to the possible changes may be made in the confirmation sheet occasionally; an up-to-date confirmation sheet can be downloaded from GENERALPLUS web site at <http://www.generalplus.com>. The following checklist is an abstract of confirmation sheet for each series.

1.2 Programming Check List

Configuration before the programming starts:

1. IO Direction, Attribution, Data default set 00h, as floating status. If not used, set input low status to reduce current.(PortA, PortB, PortC and PortD Attribution:0, Direction:0, Data:1)
2. P_IO_Green_Ctrl default is in disabled status. User must enable to reduce IO current.
3. P_SYSTEM_Green_Ctrl default is in disabled status. User must enable to reduce system current.
4. LVD function default is in disabled status. If enabled, operating current will increase 100uA.
5. PD[5] function support input high/low and output low function, but not support Output high.
6. IC waking up from sleep mode needs a little stable time to be stabilized; therefore, CPU will start running after a little stable time. User can refer to the 1.3 Wakeup stable time Differences between EV chip and GPL812P01A table for more information about EV and GPL812P01A stable time difference.
7. In EV chip, user uses Timer Base's high frequency as wakeup source; the wakeup frequency will be incorrect. The wakeup frequency source will occur in-expect wakeup frequency.
8. User only uses Real chip GPL812P01A wakeup source to get the correct wakeup frequency.
9. In GPL812P01A.inc, it may define (GPL812P01Ax_Mode) a mode for EV emulation or OTP programming which will be indicated at location 0xFFFF7 (0x1FF7), where EV emulation = 0x02 or OTP programming = 0x01.

1.3 Wakeup Stable Time Differences between EV Chip and GPL812P01A

Item	EV chip	GP812P01A
	losc 4M	losc 4M
Wakeup stable time ($\pm 5\%$)	16.5ms	600us
Wakeup source TBH 128Hz	43.1Hz*	128Hz
Wakeup source TBH 256Hz	51.2Hz*	256Hz
Wakeup source TBH 512Hz	56.8Hz*	512Hz
Wakeup source TBH 1024Hz	85.3Hz*	1024Hz

Note: Wakeup stable time causes certain wakeup frequency delay; therefore, wakeup source occurrence is inconsistent with the actual wakeup frequency (Hz). Refer to the above table for more information about the differences. The correct wakeup frequency can only be obtained from the wakeup source of GPL812P01A real chip.

2 INTRODUCTION

2.1 General Description

GPL812P01A, a unique designed CMOS 8-bit microprocessor by Generalplus, integrates RAM, OTPROM, I/Os, an interrupt controller, and an automatic display controller/driver in a small device. The extraordinary features are capable of operating in low voltage range from 2.0V ~ 3.6V. This device is suitable for many application fields such as remote control, timepiece and other LCD related products battery application.

GPL812P01A EMU board use GPL812V00A(EV) and GPL812P01A(LCM) to emulate GPL812P01A. GPL812V00A(EV)'s feature include 192 bytes working RAM, 8K bytes ROM, 22 I/Os, 8 internal interrupt sources and 2 external interrupt sources. GPL812P01A(LCM) is use to emulate LCD function, include 4 commons and 16 segments, forming a maximum of 64 dots LCD resolution.

2.2 Features

- 8-bit high performance processor
- 192-byte SRAM
- 8K-byte OTPROM
- 16-byte DPRAM
- 4MHz RC oscillator for system operation
- 32768Hz internal oscillator and 32768Hz external crystal circuit for time-base.
- Low operating voltage: 2.0V ~ 3.6V @ 4MHz
- Low standby current, $I_{STBY} < 1\mu A$ @ 3.6V@25°C
- General-purpose I/O pins: 22 pins, most of them shared with LCD segments
 - PA[7:0]
 - PB[3:0]
 - PC[1:0], PC[7:4]
 - PD[3:2], PD[5], PD[7]
- LCD configurations: 8 commons (max) x 12 segments, 4 commons x 16 segments (max).
- LCD bias: 1/3, 1/4
- LCD duty: 1/4, 1/5, 1/6, 1/8
- VCLD: 2.5V, 2.4V, 2.8V, VDD
- Frame rate: 85Hz.
- One 8-bit re-loadable timer and counter and one 12-bit re-loadable timer and counter
- Watchdog function
- 10 interrupt sources: TM0O_INT, TM1O_INT, CPUDiv1KHz_INT, CPUDiv4KHz_INT, CPUDiv32KHz_INT, CPUDiv2MHz_INT, TBH_INT, TBL_INT, EXT_INT1, and EXT_INT0
- Wakeup sources: key change-PortB[3:0], PortC[7:4,1:0], PortD[7,5,3:0], 32K TimeBase (H) (128Hz,256Hz,512Hz,1024Hz), 32K Time Base (L) (2Hz,4Hz,8Hz,16Hz)
- Internal regulator for system operation
- Built-in Low voltage reset to avoid system runaway from abnormal operation at low voltage
- Built-in RFC function

Note1: Time base low frequency: 2Hz, 4Hz, 8Hz, 16Hz

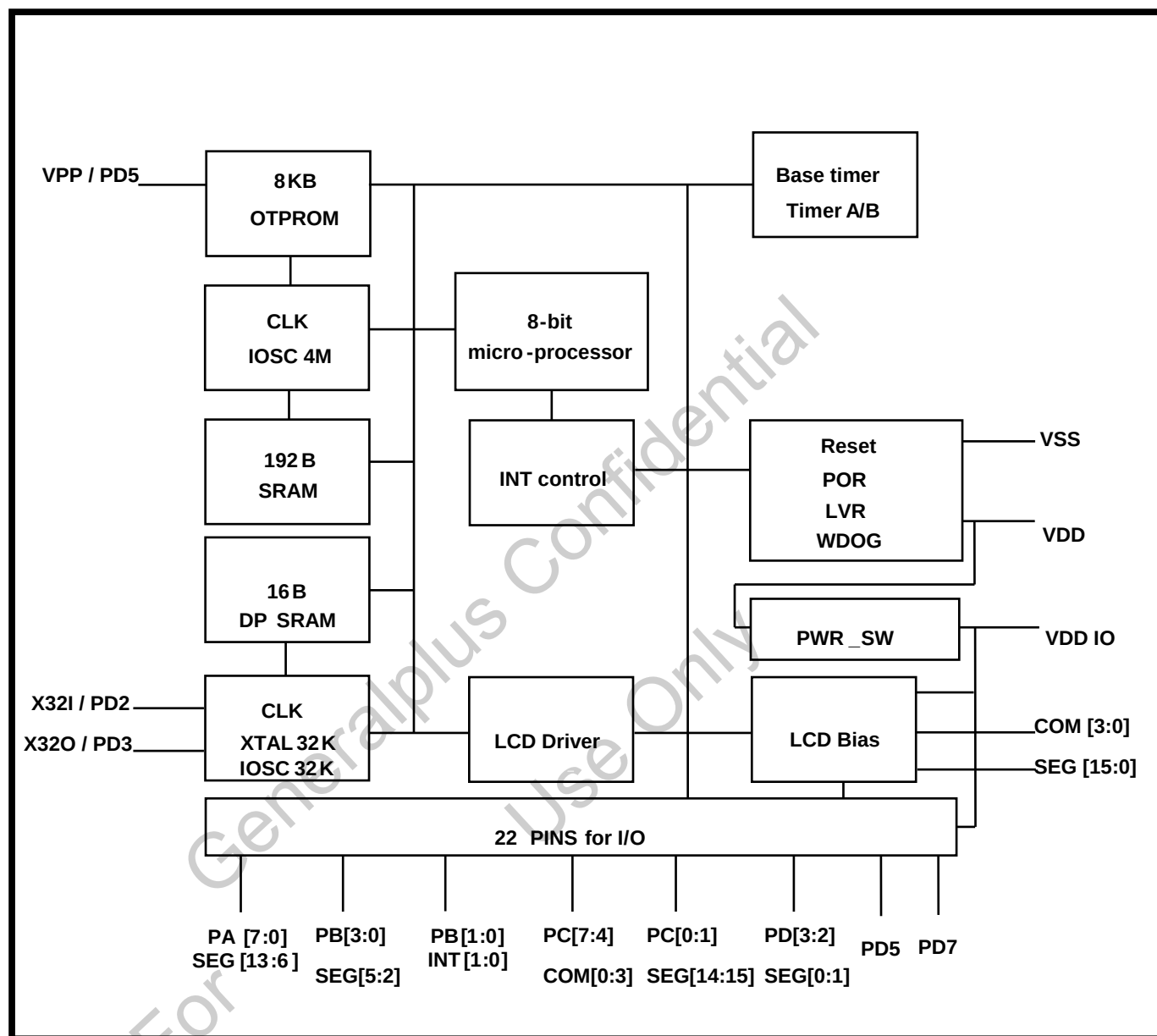
Note2: Time base high frequency: 128Hz, 256Hz, 512Hz, 1024Hz

Note3: This feature list is for user preliminary reference only. Refer to the respective data sheet for more details.

Note4: When system wakes up from sleep mode, it requires a time period to become stable and thus, CPU should wait the time period before start running the program.

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2.3 Block Diagram



2.4 Differences between Real Chip and EMU Chip

The EMU board is designed for GP812V00A and GPL812P01A, where GP812V00A is to emulate all functions except LCD; GPL812P01A acts as an LCM for LCD emulation. Registers and data access is accomplished through GP812V00A's PC[7:6] and GPL812P01A's PD[7,5].

Item		GPL812P01A EMU	GPL812P01A
		(EMU Chip + LCM)	(OTP)
Operating Voltage		2.0 – 3.6 V	2.0 – 3.6 V
ROM		8K bytes	8K bytes
RAM		192 bytes	192 bytes
DPRAM		16 bytes (LCM)	16 bytes
COM Max.		4 (LCM)	4
SEG Max.		16 (LCM)	16
LCD Dot Max		64 (LCM)	64
LCD Duty		1/4, 1/5, 1/6, 1/8 (LCM)	1/4, 1/5, 1/6, 1/8
LCD Bias		1/3, 1/4 (LCM)	1/3
VLCD		VDD, 2.8V, 2.5V, 2.4V (LCM)	VDD, 2.8V, 2.5V, 2.4V
LVD		2.1V / 2.4V	2.1V / 2.4V / 2.7V
Pure IO		PB[3:2]	X
Share IO		PA[7:0]	PA[7:0]
		PB[1:0]	PB[3:0]
		PC[7:4]	PC[7:4]
		PC[1:0]	PC[1:0]
		PD[3:2]	PD[3:2]
		PD [7]	PD [7]
RFC		V (EV)	V
Option	LVR	Always enable	Always enable
	System Clock Source	Internal ROSC 4M	Internal ROSC 4M
	32K Clock Source	Code (R30K/X32K)	Code (R30K/X32K)
	Watchdog	Code (Enable/Disable)	Code (Enable/Disable)
	Security	Code (Enable/Disable)	Code (Enable/Disable)

Address	Function	Must be set via LCM in EV simulation	Direct Setting in EV Simulation supported	Note
0x0048	P_IOA_SEGCOM_Ctrl	V	-	
0x0049	P_IOB_SEGCOM_Ctrl	V	-	
0x004B	P_IOB_LCDPORT_Ctrl	V	-	
0x0034	P_IOC_SEGCOM_Ctrl	V	-	
0x0035	P_IOD_LCDPORT_Ctrl	V	-	

Address	Function	Must be set via LCM in EV simulation	Direct Setting in EV Simulation supported	Note
0x0032	P_LCD_Ctrl	V	-	
0x004A	P_IOA_LCDPORT_Ctrl	V	-	
0x004C	P_IOC_LCDPORT_Ctrl	V	-	
0x0050 ~ 0x005F	LCD display buffer	V	-	
0x0013	P_INT_Ctrl0	-	V	Bit7/5 must be filled 0.
0x0018	P_INT_Ctrl2	-	V	Bit7/5 must be filled 0.
0x0022	P_TIMER_Timer1_Ctrl	-	V	Bit0 must be filled 0.
0x002A	P_CLK_32768_En	-	V	Bit5/2/1/0 must be filled 0.
0x0033	P_LVD_Ctrl	-	V	Bit6 Setting is invalid in EV (LVD=2.7V can't simulate)

2.5 Application Fields

- Remote controller with LCD ability
- LCD controller for home appliance
- Watch

3 CODE OPTION

3.1 Introduction

User can set code option in G+IDE/project settings/body option. The GPL812P01A LVR status is always enabled. User can set the parameters: watchdog, 32K clock source: IOSC/XTAL; if XTAL is selected, the X32O and PD3 is shared each other; also, X32I and PD2 is shared. User can enable or disable the security function as needed.

3.2 Option Settings

1. Watchdog: Disabled/Enabled
2. 32KHz clock source: IOSC/XTAL
4. Security: Disabled/Enabled

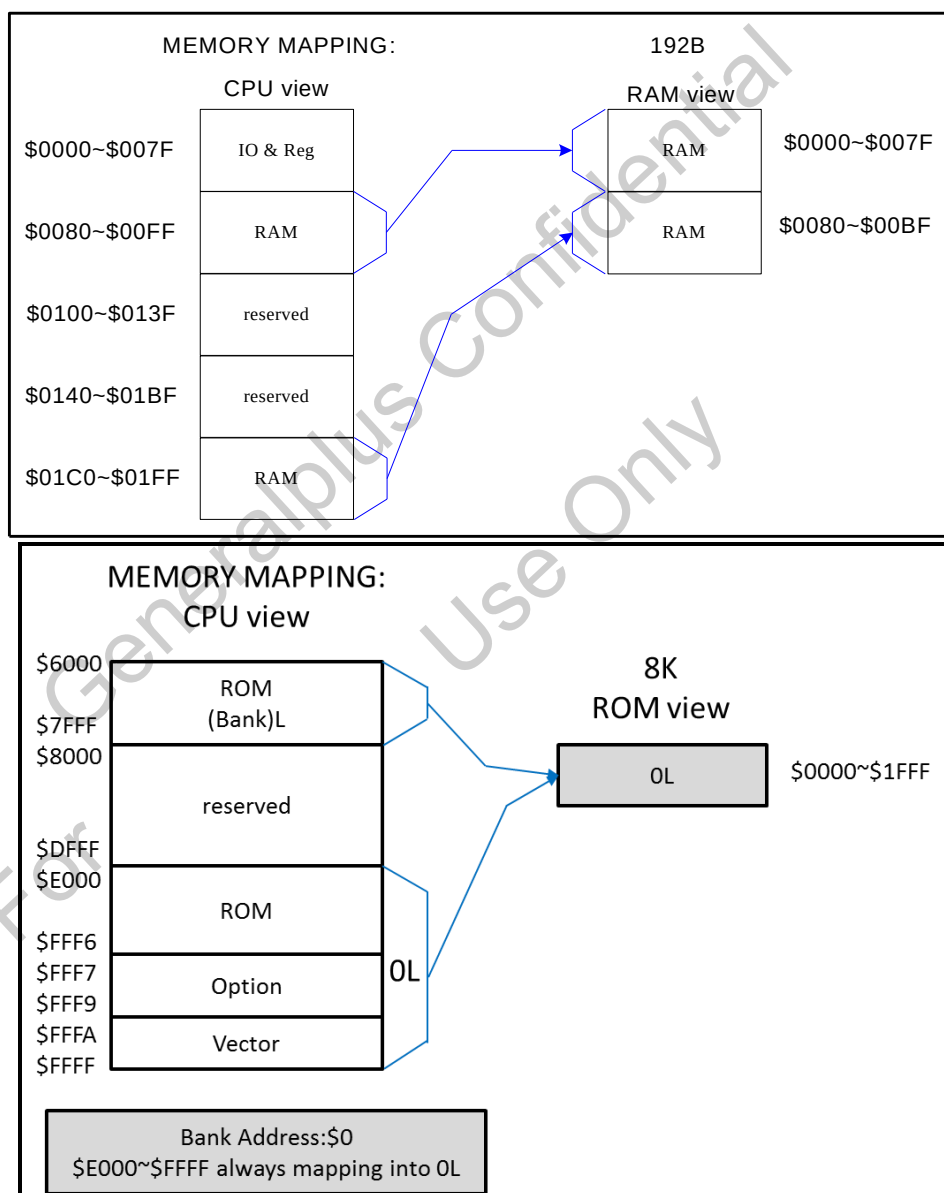
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4 MEMORY MAPPING

4.1 Introduction

GPL812P01A includes 192 bytes SRAM and various 16 bytes LCD DPRAM and 8KB OTP ROM sizes. The maximum address of CPU view is 8K bytes (0x6000~0x7FFF), and therefore, if memory size is larger than CPU view, the common area mapping to different memory address is used. The CPU view 0xE000~0xFFFF is always mapped to bank0, the first 8K bytes address.

4.2 GPL812P01A Memory Mapping



In GPL812P01A,

User's program interrupt vector: 0xFFFFA ~ 0xFFFFF

Stack area located at 0x01C0 ~ 0x01FF.

The 0xE000 ~ 0xFFFF is mapping to Bank0 0x6000 ~ 0x7FFF, ROM View: 0x0000 ~ 0x1FFF

0xFFFF7 ~ 0xFFFF9, reserved for Generalplus use only, are not allowed to be used by users.

4.3 0x003F, P_SRAM_Ctl

P_SRAM_Ctl			0x003F			SRAM Control Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	page5	-	-	-	page1	page0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:6],[4:2]	-	-	Reserved	-
[5],[1:0]	Page0/1/5	R/W	SRAM protection enabling selection If set "1", SRAM becomes read only for protection purpose. page(x) with address mapping table page0 = 0x0080~0x00BF page1 = 0x00C0~0x00FF page5 = 0x01C0~0x01FF	0: SRAM protection disabled 1: SRAM protection enabled

5 IO CONFIGURATION

5.1 Introduction

GPL812P01A has 22 general IO pins which are grouped into four parts: PortA[7:0], PortB[3:0], PortC[7:4], PortC[1:0], PortD[3:2], PortD[5], and PortD[7]. All of these IO pins can be configured as input or output function. And some pins are shared with special functions, e.g. LCD-segment, interrupt, X32I, X32O, VPP ...etc.

Note:

All IOs' default state is floating state. Don't let it be at floating state before entering sleep mode. We suggest user set IO as Input with pull-low state when program starts running.

5.2 I/O Port Shared with Other Function List

PortA								
Bit	7	6	5	4	3	2	1	0
Special	Seg13	Seg12	Seg11	Seg10	Seg9	Seg8	Seg7	Seg6
Function	COM7	COM6						

Note: PortA[7:0] can be configured as LCD function through setting P_IOA_LCDPORT_Ctrl

PortA[7:6] can be configured as LCD segment[13:12]/common[7:6] through setting P_IOA_SEGCOM_Ctrl

PortB								
Bit	7	6	5	4	3	2	1	0
Special	-	-	-	-	Seg5	Seg4	Seg3	Seg2
Function	-	-	-	-	COM3	COM2	COM	COM0
	-	-	-	-	PB3	PB2	EXT_INT1	EXT_INT0
	-	-	-	-	Key wake-up	Key wake-up	Key wake-up	Key wake-up

Note: PortB[3:0] can be configured as LCD function through P_IOB_LCDPORT_Ctrl settings.

PortB[3:0] can be configured as LCD segment[5:2]/common[3:0] through P_IOB_SEGCOM_Ctrl settings.

PortC								
Bit	7	6	5	4	3	2	1	0
	Seg5	Seg4	Seg3	Seg2	-	-	Seg15	Seg14
Special	COM0	COM1	COM2	COM3	-	-	COM4	COM5
Function	RFCI	RFCO	RFCO	RFCO	-	-	-	-
	Key wake-up	Key wake-up	Key wake-up	Key wake-up	-	-	Key wake-up	Key wake-up

Note: PortC[7:4,1:0] can be configured as LCD function through P_IOC_LCDPORT_Ctrl settings.

PortC[7:4] can be configured as LCD segment[5:2]/common[0:3] through P_IOC_SEGCOM_Ctrl settings.

PortC[1:0] can be configured as LCD segment[15:14]/common[4:5] through P_IOC_SEGCOM_Ctrl settings.

PortC[7:4] can be enabled as RFC function through P_RFC_Ctrl settings.

Don't enable LCD segment and RFC function at the same time.

PortD								
Bit	7	6	5	4	3	2	1	0
Special Function	-	-	-	-	Seg0	Seg1	-	-
	-	-	VPP	-	X32O	X32I	-	-
	Key wakeup		Key wakeup		Key wakeup	Key wakeup	-	-

Note: PortD[3:2] can be configured as LCD segment[0:1] through setting P_IOD_LCDPORT_Ctrl setting.

PortD[3:2] can be opted as 32768Hz crystal connection pin by code option.

PortD[5] shares with VPP; it results PD[5] function only NOT supporting Output high.

5.3 0x0017, P_SYSTEM_Green_Ctrl

P_SYSTEM_Green_Ctrl			0x0017		System Green mode Control Register			
Bit	7	6	5	4	3	2	1	0
Function	-	Green Mode	-	-	-	-	-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	-	-	Reserved	-
6	Green Mode	W	Set System Green mode.	0: disabled 1: enabled
[5:0]	-	-	Reserved	-

5.4 0x002C, P_IO_Strobe_En

After setting Strobe function enabled, PortB(0x002D), PortC(0x002E), and Port D(0x002F) can be used as normal I/O or LCD strobe function.

P_IO_Strobe_En			0x002C		Strobe Control Register			
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	-	-	-	StrobeEn
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:1]	-	-	Reserved	-
0	StrobeEn	R/W	Strobe function control register.	0 = Disabled 1 = Enabled

5.5 PortA Control Register Description

The PortA can be several formats via I/O register configuration. Please refer to the following table for more information about register configuration and I/O format.

Attribution (0x0003)	Direction (0x000B)	Data (0x0007)	Function	Description
0	0	0	Floating	General purpose I/O function
0	0	1	Input with pull-low	
0	1	0	Output low	
0	1	1	Output high	
1	0	0	Input with floating	
1	0	1	Input with pull-high	
1	1	0	Output high	
1	1	1	Output low	

5.6 0x000B, P_IO_PortA_Dir

P_IO_PortA_Dir			0x000B			IOA Direction control Register		
Bit	7	6	5	4	3	2	1	0
Function	IOADIR							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IOADIR	R/W	This control register sets the direction of I/O PortA.	0: input 1: output

5.7 0x0003, P_IO_PortA_Attrib

P_IO_PortA_Attrib			0x0003			IOA Attribute Register		
Bit	7	6	5	4	3	2	1	0
Function	IOAAttrib							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IOAAttrib	R/W	PortA attribution Control Write 1 to corresponding bit to set GPIO attribution.	Refer to PortA register configuration and I/O format Table

5.8 0x0007, P_IO_PortA_Data

P_IO_PortA_Data			0x0007			IOA Data Register		
Bit	7	6	5	4	3	2	1	0
Function	IOADData							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IOADData	R/W	This control register sets the data of I/O PortA.	Output mode Read: read pin current status

Bit	Function	Type	Description	Condition
				Write: set IO status Input mode Read: read pin current status Write: not change IO status

5.9 0x000C, P_IO_PortA_Buffer

P_IO_PortA_Buffer			0x000C			IOA Pin Buffer		
Bit	7	6	5	4	3	2	1	0
Function	IOABUF							
Default	X	X	X	X	X	X	X	X

Bit	Function	Type	Description	Condition
[7:0]	IOABUF	R	PortA Buffer Register Programmers can read this register to get PortA previous setting status.	Read P_IO_PortA_Data previous setting status.

5.10 0x004A, P_IOA_LCDPORT_Ctrl

The register can set PortA as normal IO or LCD function

P_IOA_LCDPORT_Ctrl			0x004A			PortA function select register		
Bit	7	6	5	4	3	2	1	0
Function	IOA_LCDEn_7	IOA_LCDEn_6	IOA_LCDEn_5	IOA_LCDEn_4	IOA_LCDEn_3	IOA_LCDEn_2	IOA_LCDEn_1	IOA_LCDEn_0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	IOA_LCDEn_[7:0]	R/W	IOA program option for IO/LCD function.	0: IO 1: LCD function

5.11 0x0048, P_IOA_SEGCOM_Ctrl

The register can set PortA as Segment or Common function control.

P_IOA_COMSEG_Ctrl			0x0048			PortA Segment/Common select register		
Bit	7	6	5	4	3	2	1	0
Function	IOA_SEGCOM_7	IOA_SEGCOM_6	-	-	-	-	-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:6]	IOA_SEGCOM_[7:6]	R/W	IOA program option for LCD segment/common function.	0 =segment output[13:12] 1 =common output[7:6]
[5:0]	-	-	Reserved	-

5.12 PortB Control Register Description

The PortB can be several formats via I/O register configuration. User should refer to the following table for more information about register configuration and I/O format.

Attribution (0x0004)	Direction (0x0000)	Data (0x0008)	Function	Description
0	0	0	Floating	General purpose I/O function
0	0	1	Input with pull-low	
0	1	0	Output low	
0	1	1	Output high	
1	0	0	Input with float	
1	0	1	Input with pull-high	
1	1	0	Output high	
1	1	1	Output low	

5.13 0x0000, P_IO_PortB_Dir

P_IO_PortB_Dir		0x0000				IOB Direction control Register			
Bit	7	6	5	4	3	2	1	0	
Function	-	-	-	-	IOBDIR				
Default	0	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	IOBDIR	R/W	This control register sets the direction of I/O PortB.	0: Input 1: Output

5.14 0x0004, P_IO_PortB_Attrib

P_IO_PortB_Attrib		0x0004				IOB Attribute Register			
Bit	7	6	5	4	3	2	1	0	
Function	-	-	-	-	IOBAttrib				
Default	0	0	0	0	0	0	0	0	0

R/O read data value 0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	IOBAttrib	R/W	PortB attribution Control Write 1 to corresponded bit to set GPIO attribution.	Refer to PortB register configuration and I/O format Table

5.15 0x0008, P_IO_PortB_Data

P_IO_PortB_Data			0x0008			IOB Data Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	IOBData			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	IOBData	R/W	This control register sets the data of I/O PortB.	Output mode Read: read pin current status Write: set IO status Input mode Read: read pin current status Write: not change IO status

5.16 0x000D, P_IO_PortB_Buffer

P_IO_PortB_Buffer			0x000D			IOB Pin Buffer Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	IOBBUF			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	IOBBUF	R	PortB Buffer Register Programmers can read this register to get PortB previous setting status.	Read P_IO_PortB_Data previous setting status.

5.17 0x002D, P_IO_PortB_Strobe_En

P_IO_PortB_Strobe_En			0x002D			IOB Strobe control Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	STRB3	STRB2	STRB1	STRB0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	STRB[3:0]	R/W	Port B strobe control register To enable P_IO_PortB_Strobe_En, need enable P_IO_Strobe_En(\$0x002c b.0) first.	0= disable IOB strobe function 1=Synchronized with LCD Strobe Waveform.

5.18 0x004B, P_IOB_LCDPORT_Ctrl

The register can set PortB as normal IO or LCD function.

P_IOB_LCDPORT_Ctrl				0x004B				PortB function select register			
Bit	7	6	5	4	3	2	1	0			
Function	-	-	-	-	IOB_LCDEn_3	IOB_LCDEn_2	IOB_LCDEn_1	IOB_LCDEn_0			
Default	0	0	0	0	0	0	0	0			

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	IOB_LCDEn_3 [3:0]	R/W	IOB program option for IO or LCD function.	0 = IO 1 = LCD function

5.19 0x0049, P_IOB_SEGCOM_Ctrl

The register can set PortB as Segment or Common function control.

P_IOB_COMSEG_Ctrl				0x0049				PortB Segment/Common select register			
Bit	7	6	5	4	3	2	1	0			
Function	-	-	-	-	IOC_SEGCOM_3	IOC_SEGCOM_2	IOC_SEGCOM_1	IOC_SEGCOM_0			
Default	0	0	0	0	0	0	0	0			

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	IOB_SEGCOM_3 [3:0]		IOB program option for segment/common of LCD function.	0: LCD segment [5:2]. 1: LCD common [3:0].

5.20 PortC Control Register Description

The PortC can be several formats via I/O control and direct register configuration. Please refer to the following table for more information about register configuration and I/O format.

Attribution (0x0005)	Direction (0x0001)	Data (0x0009)	Function	Description
0	0	0	Floating	General purpose I/O function
0	0	1	Input with pull-low	
0	1	0	Output low	
0	1	1	Output high	
1	0	0	Input with float	
1	0	1	Input with pull-high	
1	1	0	Output high	
1	1	1	Output low	

5.21 0x0001, P_IO_PortC_Dir

P_IO_PortC_Dir				0x0001				IOC Direction control Register			
Bit	7	6	5	4	3	2	1	0			
Function	IOCDIR [7:4]				-	-	IOCDIR [1:0]				
Default	0	0	0	0	0	0	0	0			

Bit	Function	Type	Description	Condition
[7:4]	IOCDIR [7:4]	R/W	This control register sets the direction of I/O PortC.	0: Input 1: Output
[3:2]	-	-	Reserved	-
[1:0]	IOCDIR [1:0]	R/W	This control register sets the direction of I/O PortC.	0: Input 1: Output

5.22 0x0005, P_IO_PortC_Attrib

P_IO_PortC_Attrib			0x0005		IOC Data Register			
Bit	7	6	5	4	3	2	1	0
Function	IOCAAttrb [7:4]				~	~	IOCAAttrb [1:0]	
Default	0	0	0	0	~	~	0	0

Bit	Function	Type	Description	Condition
[7:4]	IOCAAttrb [7:4]	R/W	PortC attribution Control Write 1 to corresponded bit to set GPIO attribution.	Refer to PortC register configuration and I/O format Table
[3:2]	-	-	reserved	-
[1:0]	IOCData[1:0]	R/W	PortC attribution Control Write 1 to corresponded bit to set GPIO attribution.	Refer to PortC register configuration and I/O format Table

5.23 0x0009, P_IO_PortC_Data

P_IO_PortC_Data			0x0009		IOC Data Register			
Bit	7	6	5	4	3	2	1	0
Function	IOCData [7:4]				~	~	IOCData [1:0]	
Default	0	0	0	0	~	~	0	0

Bit	Function	Type	Description	Condition
[7:4]	IOCData[7:4]	R/W	This control register sets the data of I/O PortC.	Output mode: Read: read pin current status Write: set IO status Input mode: Read: read pin current status Write: not change IO status
[3:2]	-	-	reserved	-
[1:0]	IOCData[1:0]	R/W	This control register sets the data of I/O PortC.	Output mode: Read: read pin current status Write: set IO status Input mode: Read: read pin current status Write: not change IO status

5.24 0x000E, P_IO_PortC_Buffer

P_IO_PortC_Buffer			0x000E			IOC Pin Buffer Register		
Bit	7	6	5	4	3	2	1	0
Function	IOCBUF[7:4]				~	~	IOCBUF[1:0]	
Default	0	0	0	0	~	~	0	0

Bit	Function	Type	Description	Condition
[7:4]	IOCBUF[7:4]	R	PortC Buffer Register Programmers can read this register to get PortC's previous setting status.	Read P_IO_PortC_Data previous setting status.
[3:2]	-	-	reserved	-
[1:0]	IOCBUF[1:0]	R	PortC Buffer Register Programmers can read this register to get PortC's previous setting status.	Read P_IO_PortC_Data previous setting status.

5.25 0x004C, P_IOC_LCDPORT_Ctrl

The register can set PortC as normal IO or LCD function

P_IOC_LCDPORT_Ctrl			0x004C			PortC function select register		
Bit	7	6	5	4	3	2	1	0
Function	IOC_LCDEn_7	IOC_LCDEn_6	IOC_LCDEn_5	IOC_LCDEn_4	~	~	IOC_LCDEn_1	IOC_LCDEn_0
Default	0	0	0	0	~	~	0	0

Bit	Function	Type	Description	Condition
[7:4]	IOC_LCDEn_7 [7:4]	R/W	IOC program option for IO/LCD function.	0: IO 1: LCD function
[3:2]	-	-	reserved	-
[1:0]	IOC_LCDEn_1 [1:0]	R/W	IOC program option for IO/LCD function	0: IO 1: LCD function

5.26 0x0034, P_IOC_SEGCOM_Ctrl

The register can set PortC as Segment or Common function control.

P_IOC_SegCom_Ctrl			0x0034			PortC Segment/Common select register		
Bit	7	6	5	4	3	2	1	0
Function	IOC_SEGCOM_7	IOC_SEGCOM_6	IOC_SEGCOM_5	IOC_SEGCOM_4	R/O	R/O	IOC_SEGCOM_1	IOC_SEGCOM_0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	IOC_SEGCOM_7 [7:4]	R/W	IOC program option for segment/common of LCD function.	0: LCD segment [5:2]. 1: LCD common [0:3].
[3:2]	-	-	reserved	-
[1:0]	IOC_SEGCOM_1 [1:0]	R/W	IOC program option for segment/common of LCD function.	0: LCD segment [15:14]. 1: LCD common [4:5].

5.27 0x002E, P_IO_PortC_Strobe_En

P_IO_PortC_Strobe_En			0x002E			IOC Strobe control Register		
Bit	7	6	5	4	3	2	1	0
Function	STRC7	STRC6	STRC5	STRC4	-	-	STRC1	STRC0
Default	0	0	0	0	-	-	0	0

Bit	Function	Type	Description	Condition
[7:4]	STRC[7:4]	R/W	Port C strobe control register To enable P_IO_PortC_Strobe_En, we should first enable P_IO_Strobe_En(\$0x002c b.0).	0: disable PortC[7:4] strobe function 1: Synchronized with LCD Strobe Waveform
[3:2]	-	-	reserved	-
[1:0]	STRC[1:0]	R/W	Port C strobe control register To enable P_IO_PortC_Strobe_En, we should first enable P_IO_Strobe_En(\$0x002c b.0).	0: disable PortC[1:0] strobe function 1: Synchronized with LCD Strobe Waveform

5.28 PortD Control Register Description

The PortD can be several formats via I/O register configuration. User can refer to the following table for more information about register configuration and I/O format.

Attribution (0x0006)	Direction (0x0002)	Data (0x000A)	Function	Description
0	0	0	Floating	General purpose I/O function
0	0	1	Input with pull-low	
0	1	0	Output low	
0	1	1	Output high	
1	0	0	Input with float	
1	0	1	Input with pull-high	
1	1	0	Output high	
1	1	1	Output low	

Note: PD[5] function supports input high/low and output low function, but does not support Output high.

5.29 0x0002, P_IO_PortD_Dir

P_IO_PortD_Dir			0x0002			IOD Direction control Register		
Bit	7	6	5	4	3	2	1	0
Function	IODDIR	-	IODDIR	-	IODDIR		-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	IODDIR[7]	R/W	This control register sets the direction of I/O PortD[7].	0: Input 1: Output

Bit	Function	Type	Description	Condition
6	-	-	Reserved	-
5	IODDIR[5]	R/W	This control register sets the direction of I/O PortD[5]. The PD[5] function supports input high/low and output low function, but does not support output high.	0: Input 1: Output
4	-	-	Reserved	-
[3:2]	IODDIR[3:0]	R/W	This control register sets the direction of I/O PortD[3:0].	0: Input 1: Output
[1:0]	-	-	Reserved	-

5.30 0x0006, P_IO_PortD_Attrib

P_IO_PortD_Attrib			0x0006			IOD Data Register		
Bit	7	6	5	4	3	2	1	0
Function	IODDIR	-	IODDIR	-	IODAttrib		-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	IODAttrib[7]	R/W	PortD Buffer Register Write "1" to corresponded bit to set GPIO High/Low Status. Programmers can read this register to get PortD[7] setting status.	Refer to PortD register configuration and I/O format Table
6	-	-	Reserved	-
5	IODAttrib[5]	R/W	PortD Buffer Register Write "1" to corresponded bit to set GPIO High/Low Status. Programmers can read this register to get PortD[5] setting status.	Refer to PortD register configuration and I/O format Table
4	-	-	Reserved	-
[3:2]	IODAttrib[3:2]	R/W	PortD Buffer Register Write "1" to corresponded bit to set GPIO High/Low Status. Programmers can read this register to get PortD[3:2] setting status.	Refer to PortD register configuration and I/O format Table
[1:0]	-	-	Reserved	-

5.31 0x000A, P_IO_PortD_Data

P_IO_PortD_Data			0x000A			IOD Data Register		
Bit	7	6	5	4	3	2	1	0
Function	IODData	-	IODData	-	IODData		-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	IODData[7]	R/W	This control register sets the data of I/O PortD[7].	Output mode Read: read pin current status Write: set IO status Input mode Read: read pin current status Write: not change IO status
6	-	-	Reserved	-
5	IODData[5]	R/W	This control register sets the data of I/O PortD[5].	Output mode Read: read pin current status Write: set IO status Input mode Read: read pin current status Write: not change IO status
4	-	-	Reserved	-
[3:2]	IODData[3:2]	R/W	This control register sets the data of I/O PortD[3:2].	Output mode Read: read pin current status Write: set IO status Input mode Read: read pin current status Write: not change IO status
[1:0]	-	-	Reserved	-

5.32 0x000F, P_IO_PortD_Buffer

P_IO_PortD_Buffer			0x000F			IOD Pin Buffer Register		
Bit	7	6	5	4	3	2	1	0
Function	IODBUF	-	IODBUF	-	IODBUF		-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	IODBUF[7]	R	Programmers can read this register to get PortD.b7 setting status previous.	Read P_IO_PortD_Data previous setting status.
6	-	-	Reserved	-
5	IODBUF[5]	R	Programmers can read this register to get PortD.b5 previous setting status.	Read P_IO_PortD_Data previous setting status.
4	-	-	Reserved	-
[3:2]	IODBUF[3:2]	R	Programmers can read this register to get PortD[3:2] previous setting.	Read P_IO_PortD_Data previous setting status.
[1:0]	-	-	Reserved	-

Note: PD5 for GPL812P02A / GPL81202A Used Only.

5.33 0x0011, P_IO_Green_Ctrl

P_IO_Green_Ctrl			0x0011			IO Green mode Control Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	Green Mode	-	-	-	-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:6]	-	-	Reserved	-
[5]	Green Mode	W	Set IO Green mode.	0: disable. 1: enable.
[1:0]	-	-	Reserved	-

5.34 0x0035, P_IOD_LCDPORT_Ctrl

The register can set PortD as normal IO or LCD segment.

P_IOD_LCDPORT_Ctrl			0x0035			PortD function select register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	IOD_SEGEn 3	IOD_SEGEn 2	-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:2]	IOC_SEGEn [3:2]	R/W	IOD code option for IO/LCD segment function.	0: IO 1: LCD segment[0:1]
[1:0]	-	-	Reserved	-

5.35 0x002F, P_IO_PortD_Strobe_En

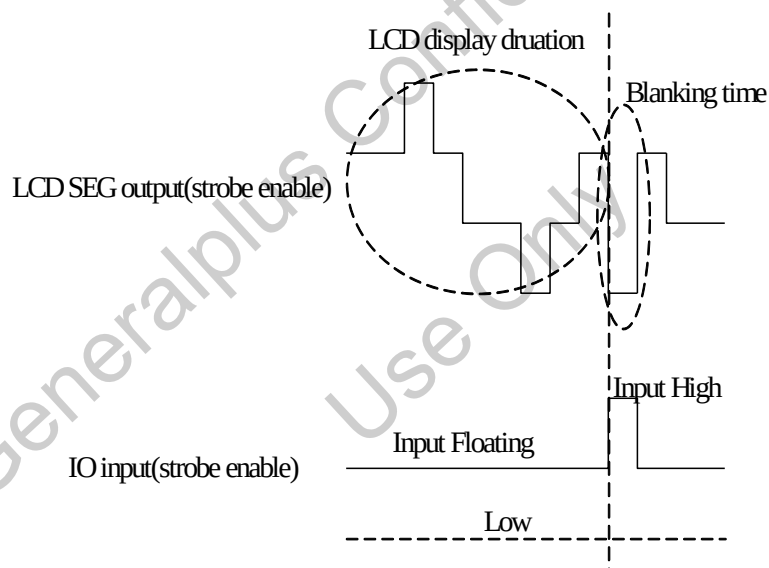
P_IO_PortD_Strobe_En			0x002F			IOD Strobe control Register		
Bit	7	6	5	4	3	2	1	0
Function	STRD7	-	STRD5	-	STRD3	STRD2	-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	STRD7	R/W	PortD[7] strobe control register To enable P_IO_PortD_Strobe_En, we should enable P_IO_Strobe_En(\$0x002c b.0) first.	0: disable PortD.b7 strobe function 1: Synchronized with LCD Strobe Waveform
6	-	-	Reserved	-
5	STRD5	R/W	PortD[5] strobe control register To enable P_IO_PortD_Strobe_En, we should enable P_IO_Strobe_En(\$0x002c b.0) first.	0: disable PortD.b5 strobe function 1: Synchronized with LCD Strobe Waveform
4	-	-	Reserved	-

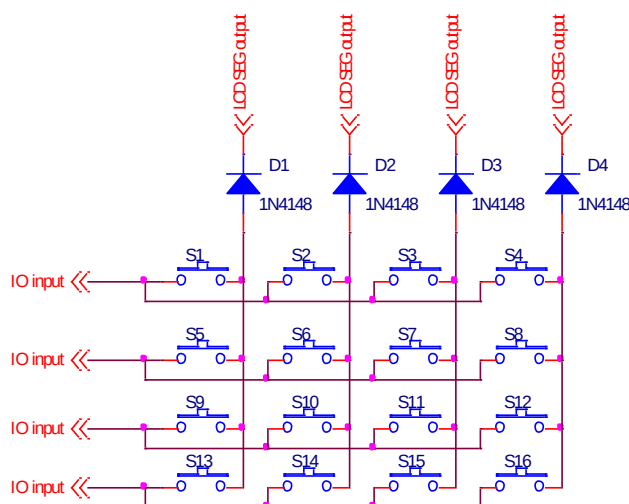
Bit	Function	Type	Description	Condition
[3:2]	STRD[3:2]	R/W	PortD[3:2] strobe control register To enable P_IO_PortD_Strobe_En, we should enable P_IO_Strobe_En(\$0x002c b.0) first.	0: disable PortD[3:2] strobe function 1: Synchronized with LCD Strobe Waveform
[1:0]	-	-	Reserved	-

5.36 IO Key Strobe and Key Scan

GPL812P01A IO strobe function is used for key change wakeup when key scan output is shared with LCD segment. The circuit shows as below. GPL812P01A IOA/IOB/IOC/IOD can program set as IO or LCD segment signal. When IO is set as LCD segment and P_IO_Strobe_En enabled, the LCD segment will output low pulse at LCD blanking time. This duration can be key change wakeup. After key change wakeup, user needs to disable strobe function and can change these LCD segment as normal IO and process key scan service. If user does not use LCD segment as key scan output, IO strobe function does not need to be enabled.



Here, LCD SEG output is IOA or IOC, set as LCD SEG mode. IO input is IOA/IOB/IOC/IOD, used as normal IO.

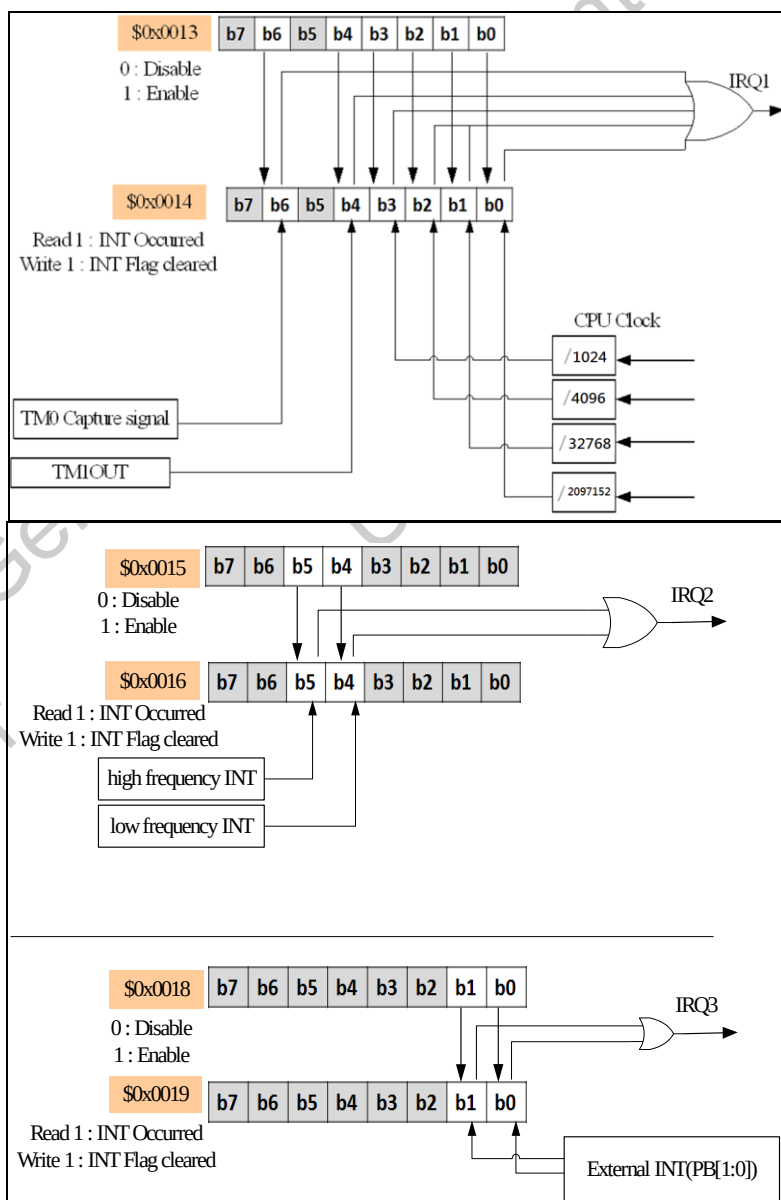


6 INTERRUPT / WATCHDOG CONFIGURATION

6.1 Introduction

GPL812P01A supports 10 interrupt sources: TM0O_INT, TM1O_INT, CPUDiv1024_INT, CPUDiv4096_INT, CPUDiv32768_INT, CPUDiv2097152_INT, TBH_INT, TBL_INT, EXT_INT1, and EXT_INT0. All interrupts can be ON/OFF separately through setting TM0O_INT, TM1O_INT, CPUDiv1024_INT, CPUDiv4096_INT, CPUDiv32768_INT, CPUDiv2097152_INT by 0x0013, setting TBH_INT and TBL_INT by 0x0015[5:4], setting EXT_INT1 and EXT_INT0 by 0x0018[1:0] control register.

If user utilizes timer count as interrupt source such as CPUDiv1024_INT, CPUDiv4096_INT, CPUDiv32768_INT, and CPUDiv2097152_INT. These timers count clock sources are from CPU Clock. It depends on Fosc clock source code option.



GPL812P01A also has a built-in watchdog function that can be enabled by code option. If watchdog function is

code option enabled and it is not cleared within 1 second, a system reset event will occur. We suggest user clear watchdog within 1 second to avoid accidental reset.

6.2 0x0013, P_INT_Ctrl0

P_INT_Ctrl0		0x0013			Interrupt Control Register			
Bit	7	6	5	4	3	2	1	0
Function	0	TM0O_INT	0	TM1O_INT	CPUDiv1K_INT	CPUDiv4K_INT	CPUDiv32K_INT	CPUDiv2M_INT
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	-	-	P_INT_Ctrl0 bit7 must be fill '0'	-
6	TM0O_INT	RW	Timer 0 overflow interrupt enabling selection	0 = Disabled 1 = Enabled
5	-	-	P_INT_Ctrl0 bit5 must be filled '0'	-
4	TM1O_INT	RW	Timer1 overflow interrupt enabling selection	0 = Disabled 1 = Enabled
3	CPUDiv1K_INT	RW	CPUCLK/1024 frequency interrupt enabling selection The flag CPUDiv1K_Clr will be set when CPUCLK/1024 frequency has done regardless CPUDiv1K_INT is enabled or disabled. We suggest user clear CPUDiv1K_Clr flag first before enabling CPUDiv1K interrupt.	0 = Disabled 1 = Enabled
2	CPUDiv4K_INT	RW	CPUCLK/4096 frequency interrupt enabling selection The flag CPUDiv4K_Clr will be set when CPUCLK/4096 frequency has done regardless CPUDiv4K_INT is enabled or disabled. We suggest user clear CPUDiv4K_Clr flag first before enabling CPUDiv4K interrupt.	0 = Disabled 1 = Enabled
1	CPUDiv32K_Hz_INT	RW	CPUCLK/32768 frequency interrupt enabling selection The flag CPUDiv32K_Clr will be set when CPUCLK/32768 frequency has done regardless CPUDiv32K_INT is enabled or disabled. We suggest user clear CPUDiv32K_Clr flag first before enabling CPUDiv32K interrupt.	0 = Disabled 1 = Enabled
0	CPUDiv2MHz_INT	RW	CPUCLK/2097152 frequency interrupt enabling selection The flag CPUDiv2M_Clr will be set when CPUCLK/2097152 frequency has done regardless CPUDiv2M_INT is enabled or disabled. We suggest user clear CPUDiv2M_Clr flag first before enabling CPUDiv2M interrupt.	0 = Disabled 1 = Enabled

Note: Clear interrupt flag before using CPUDiv1K, 4K, 32K, 2M interrupt. The interrupt will auto raise its flag. If interrupt is disabled, although interrupt flag is raised, it will not jump to Interrupt Service Routine.

6.3 0x0014, P_INT_Clear0

P_INT_Clear0		0x0014				Interrupt Control Flag Clear Register		
Bit	7	6	5	4	3	2	1	0
Function	-	TM0O_Clr	-	TM1O_Clr	CPUDiv1K_Clr	CPUDiv4K_Clr	CPUDiv32K_Clr	CPUDiv2M_Clr
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	-	-	Reserved	-
6	TM0O_Clr	RW	Timer 0 overflow interrupt flag	Read 1: Timer 0 overflow interrupt occurred. Write 1: Timer 0 overflow interrupt flag cleared.
5	-	-	Reserved	-
4	TM1O_Clr	RW	Timer 1 overflow interrupt flag	Read 1: Timer 1 overflow interrupt occurred. Write 1: Timer 1 overflow interrupt flag cleared.
3	CPUDiv1K_Clr	RW	CPUCK/1K frequency interrupt flag This flag will be set when CPUCK/1K frequency has done regardless CPUDiv1K_INT is enabled or disabled. We suggest user clear this flag first before enabling CPUDiv1K interrupt.	Read 1: CPUCK/1K frequency interrupt occurred. Write 1: CPUCK/1K frequency interrupt flag cleared.
2	CPUDiv4K_Clr	RW	CPUCK/4K frequency interrupt flag This flag will be set when CPUCK/4K frequency has done regardless CPUDiv4K_INT is enabled or disabled. We suggest user clear this flag first before enabling CPUDiv4K interrupt.	Read 1: CPUCK/4K frequency interrupt occurred. Write 1: CPUCK/4K frequency interrupt flag cleared.
1	CPUDiv32K_Clr	RW	CPUCK/32768 frequency interrupt flag This flag will be set when CPUCK/32K frequency has done regardless CPUDiv32K_INT is enabled or disabled. We suggest user clear this flag first before enabling CPUDiv32K interrupt.	Read 1: CPUCK/32K frequency interrupt occurred. Write 1: CPUCK/32K frequency interrupt flag cleared.
0	CPUDiv2M_Clr	RW	CPUCK/2M frequency interrupt flag This flag will be set when CPUCK/2097152 frequency has done regardless CPUDiv2M_INT is enabled or disabled. So we suggest user clear this flag first before enabling CPUDiv2M interrupt.	Read 1: CPUCK/2M frequency interrupt occurred. Write 1: CPUCK/2M frequency interrupt flag cleared.

6.4 0x0015, P_INT_Ctrl1

P_INT_Ctrl1			0x0015			Interrupt Control Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	TBH_INT	TBL_INT	-	-	-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:6]	-	-	Reserved	-
5	TBH_INT	R/W	32K Time Base high frequency interrupt enabling selection TBH frequency can be selected through P_TIMER_TimeBase_Sel	0 = Disabled 1 = Enabled
4	TBL_INT	R/W	32K Time Base low frequency interrupt enabling selection TBL frequency can be selected through P_TIMER_TimeBase_Sel	0 = Disabled 1 = Enabled
[3:0]	-	-	Reserved	-

6.5 0x0016, P_INT_Clear1

P_INT_Clear1			0x0016			Interrupt Flag Clear Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	TBH_Clr	TBL_Clr	-	-	-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:6]	-	-	Reserved	-
5	TBH_Clr	R/W	32K Time Base high frequency interrupt Flag	Read 1: Time Base high frequency count interrupt occurred. Write 1: Time Base high frequency count interrupt flag cleared.
4	TBL_Clr	R/W	32K Time Base low frequency interrupt Flag.	Read 1: Time Base low frequency count interrupt occurred. Write 1: Time Base low frequency count interrupt flag cleared
[3:0]	-	-	Reserved	-

6.6 0x0018, P_INT_Ctrl2

P_INT_Ctrl2			0x0018			Interrupt IO Register		
Bit	7	6	5	4	3	2	1	0
Function	0	-	0	-	-	-	EXT_INT1	EXT_INT0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7			P_INT_Ctrl2 bit7 must be fill "0"	
6	-	-	Reserved	-
5	-	-	P_INT_Ctrl2 bit5 must be fill "0"	-
[4:2]	-	-	Reserved	-
1	EXT_INT1	R/W	External INT1 interrupt enabling selection The external INT1 pin is IOB1.	0 = Disabled 1 = Enabled
0	EXT_INT0	R/W	External INT0 interrupt enabling selection The external INT0 pin is IOB0.	0 = Disabled 1 = Enabled

6.7 0x0019, P_INT_Clear2

P_INT_Clear2			0x0019		Interrupt IO Flag Register			
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	-	-	INT1_Clr	INT0_Clr
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:2]	-	-	Reserved	-
1	INT1_Clr	R/W	External INT1 Interrupt flag	Read 1: external INT1 interrupt occurred. Write 1: external INT1 interrupt flag cleared
0	INT0_Clr	R/W	External INT01 Interrupt flag	Read 1: external INT0 interrupt occurred. Write 1: external INT0 interrupt flag cleared

6.8 0x0020, P_WDT_Flag_Clear

P_WDT_Flag_Clear			0x0020		Watchdog Clear Register			
Bit	7	6	5	4	3	2	1	0
Function	WDTCLEAR							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	WDTCLEAR	W	Writing AAH to this register will clear watchdog count. Write other data to rest system.	-

Note: GENERALPLUS highly recommends user clear WDT at least every 1 second to avoid accidental reset event.

7 TIMER

7.1 Introduction

GPL812P01A includes one 8-bit timer and one 12-bit timer. The clock source comes from Fosc and Fosc divide. Initial value of the down-counter is stored in the preloaded register.

7.2 0x0021, P_TIMER_Timer0_Ctrl

P_TIMER_Timer0_Ctrl		0x0021				Timer0 Control Register		
Bit	7	6	5	4	3	2	1	0
Function	TM0ES	-	TM0CLK1	TM0CLK0	-	-	-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	TM0ES	R/W	Timer 0 enable/disable selection	0: Disabled 1: Enabled
6	-	-	Reserved	-
[5:4] [1:0]	TM0CLK	R/W	Timer 0 clock source select bits	00 = Fosc 01 = Fosc/2 10 = Fosc/4 11 = Fosc/16
[3:0]	-	-	Reserved	-

7.3 0x0022, P_TIMER_Timer1_Ctrl

P_TIMER_Timer1_Ctrl		0x0022				Timer1 Control Register		
Bit	7	6	5	4	3	2	1	0
Function	TM1ES	TM1CLKIO	TM1CLK1	TMACLK0	-	-	-	0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	TM1ES	R/W	Timer1 enabling/disabling selection	0: Disabled 1: Enabled
6	TM1CLKIO	R/W	Timer1 clock source from IO selection	0 = Timer 1 clock source setting from bit[5:4]. 1 = Timer 1 clock source from IO portC[7]. (RFC using)
[5:4] [1:0]	TM1CLK1	R/W	Timer1 clock source select bits TM0OF: clock source use Timer0 clock over flow.	00 = Fosc 01 = Fosc/4 10 = Fosc/64 11 = TM0OF
[3:1]	-	-	Reserved	-
0	-	-	P_TIMER_Timer1_Ctrl bit0 Must be filled '0'.	-

7.4 0x0023, P_TIMER_TM0Data_LB

P_TIMER_TM0Data_LB		0x0023				Timer Count Data Register		
Bit	7	6	5	4	3	2	1	0
Function	P_TIMER_TM0Data_LB							
Default	1	1	1	1	1	1	1	1

Bit	Function	Type	Description	Condition
[7:0]	TM0Data0	R/W	R: TM0 Counter 8 bits W: TM0 8 Bits Register	When counting down to 0, it will auto re-load the setting value and can issue timer overflow interrupt.

7.5 0x0025, P_TIMER_TM1Data_LB

P_TIMER_TM1Data_LB		0x0025				Timer1 Count Low Byte Register		
Bit	7	6	5	4	3	2	1	0
Function	TM1Data_LB							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	TM1Data_LB	R/W	R : TM1 Counter Low 8 bits W : TM1 Low 8 Bits Register	

7.6 0x0026, P_TIMER_TM1Data_HB

P_TIMER_TM1Data_HB		0x0026				Timer1 Count High Byte Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	TM1 High 4 Bits Register			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	TM1Data_HB	R/W	TM1 Counter high 4 bits	

7.7 0x002A, P_CLK_32768_En

P_CLK_32768_En		0x002A				C32K control Register		
Bit	7	6	5	4	3	2	1	0
Function	EN	-	0	-	-	0	0	0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7]	EN	R/W	This bit is used to control CLK32768 ON/OFF If read, reset 32K counter, read function RESET C32 Counter.	Write 1: CLK32768 enabled Write 0: CLK32768 disabled Read : Reset CLK32768 counter

Bit	Function	Type	Description	Condition
			If Write 0, disable C32K.	
[6]	-	-	Reserved	-
[5]	-	-	P_CLK_32768_En bit5 must be fill '0'	-
[4:3]	-	-	Reserved	-
2	-		P_CLK_32768_En bit2 must be fill '0'	-
1	-		P_CLK_32768_En bit1 must be fill '0'	-
0			P_CLK_32768_En bit0 must be fill '0'	

7.8 0x002B, P_TIMER_TimeBase_Sel

P_TIMER_TimeBase_Sel		0x002B				Timer control Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	32kStr_T	TBLSEL1	TBLSEL0	TBHSEL1	TBHSEL0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:5]	-	-	Reserved	-
4	32kStr_T	R/W	Strong: crystal 32KHz strong mode duration selection	0 : 2 seconds (Default) 1 : 4 seconds
[3:2]	TBLSEL[1:0]	R/W	Timebase low frequency selection	00 : 2Hz 10 : 4Hz 01 : 8Hz 11 : 16Hz
[1:0]	TBHSEL[1:0]	R/W	Timebase high frequency selection	00 : 128Hz 10 : 256Hz 01 : 512Hz 11 : 1024Hz

8 LCD CONFIGURATION

8.1 Introduction

The LCD's control and display functions are available only in GPL812P01A. The EV chip is capable of performing LCD display function via LCM (LCD module) driver because the LCM mentioned here is actually the GPL812P01A. Thus, the LCD display shown on the EV chip is what we see on GPL812P01A. For more information about the examples of EV chip's control, please refer to Section 8.4.

8.2 GPL812P01A Buffer Table

The SRAM of, \$50 - \$5F areas are reserved for LCD display buffer. Simply write the corresponding bit in display buffer to display a pattern on LCD. The following table is the mapping of LCD buffer RAM. The LCD RAM can be read and written.

	SEG0-SEG7	SEG8-SEG15
	b0-b7	b0-b7
Com0	0x50	0x51
Com1	0x52	0x53
Com2	0x54	0x55
Com3	0x56	0x57
Com4	0x58	0x59
Com5	0x5A	0x5B
Com6	0x5C	0x5D
Com7	0x5E	0x5F

8.3 0x0032, P_LCD_Ctrl

P_LCD_Ctrl		0x0032			LCD Control Register			
Bit	7	6	5	4	3	2	1	0
Function	Green Mode1	VLCDSEL		Green Mode0	LBSDT1	LBSDT0	LCDMD1	LCDMD0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7][4]	Green Mode	R/W	Reduce LCD driving to reduce current consumption	Reset Value= 11 11: disabled 10: enabled
[6:5]	VLCDSEL	R/W	VLCD Voltage Selection	00 : 2.5V 01 :2.4V 10 :2.8V 11 :VDD
[3:2]	LBSDT[1:0]	R/W	LCD bias and duty setting	00: 1/3 BIAS 1/4 DUTY 01: 1/3 BIAS 1/5 DUTY 10: 1/3 BIAS 1/6 DUTY 11: 1/4 BIAS 1/8 DUTY

Bit	Function	Type	Description	Condition
[1:0]	LCDMD[1:0]	R/W	LCD display setting	00: normal 01: all on (display black) 10: all off (display white) 11: display off. LCD signal output low state.

Note: In EV chip, LCD registers must be configured through LCM. Directly configure to EV chip is invalid.

8.4 Example Program

```

; =====
; LCD Initial Example Program
; LCD Control Register Initial
; =====
.if (GPL812P01Ax_Mode=EV_Simulation)
    LDA    #P_LCD_Ctrl
    LDY    #(D_LCDMode_Halt+D_LCDVreg_2p8V+D_LCDSetting_3Bias_4Duty)
    JSR    F_LCM_Write_OneReg
.else

    LDA    #(D_LCDMode_Halt+D_LCDVreg_2p8V+D_LCDSetting_3Bias_4Duty)
    STA    P_LCD_Ctrl
.endif

```

9 SLEEP / WAKE UP

9.1 SLEEP / WAKEUP

GPL812P01A supports sleep mode for power saving. In sleep mode, the CPU clock will stop and RAM & IO state will retain at their previous states. GPL812P01A supports sleep mode to reduce power consumption. The sleep mode includes both halt mode 1 and halt mode 2. In halt mode 1 and halt mode 2, CPU clock will stop and CLK32768 will still be operating. The difference between halt mode1 and halt mode2 is that user can further set P_LCD_Ctrl[1:0] to 11', which will turn off LCD display to reduce power consumption.

	Operating mode	Halt mode 1	Halt mode 2
CPU	✓	✗	✗
CLK32768	✓	✓	✓ (LCD display off by user setting)

There are two ways to make system wake up from halt mode:

(1) key wake-up capability: PortB[3:0], PortC[7:4,1:0], PortD[7,5,3:2]

(2) 32K TimeBase wake-up in high frequency (128Hz, 256Hz, 512Hz, 1024Hz) and low frequency (2Hz, 4Hz, 8Hz, 16Hz). When IC wakes up from sleep mode, it needs a time period becoming stable; therefore, CPU should start running a program after the time period. In EV chip, when using Timer-base high frequency for wakeup source, the wakeup frequency is not quite correct. The wakeup frequency source differs from the actual wakeup frequency. For more information about the differences, refer to Section 1.3 wakeup stable time differences between EV chip and GPL812P01A. When GPL812P01A wakes up from sleep mode, CPU will run the next instruction from where the system goes sleep. The wakeup event will not issue system reset, and therefore any initial configuration needs to be set by software.

9.2 0x0012, P_SYSTEM_Ctrl

Set up the wakeup source, before enter sleep mode.

P_SYSTEM_Ctrl	0x0012				System Control Register			
Bit	7	6	5	4	3	2	1	0
Function	SLEEP							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:0]	SLEEP	W	Write AAH to this register, CPU will stop to enter sleep mode. Write other data will cause system reset	Write AA to sleep. Write any data to system reset.

9.3 0x0030, P_WAKEUP_Ctrl

P_WAKEUP_Ctrl			0x0030			Wake-up Control Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	TBHWEN	-	TBLWEN	KEYWEN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
3	TBHWEN	R/W	32K Time Base High Frequency Wakeup Source Enable Note: (1) User can set time base high frequency by \$2B [3:2]. (2) High frequency : 128Hz, 256Hz, 512Hz, 1024Hz.	0 = Disable and clear flag 1 = Enable
2	-	-	Reserved	-
1	TBLWEN	R/W	32K Time Base Low Frequency Wakeup Source Enable Note: (1) User can set time base low frequency by \$2B [1:0]. (2) Low frequency : 2Hz, 4Hz, 8Hz, 16Hz.	0 = Disable and clear flag 1 = Enable
0	KEYWEN	R/W	Key wakeup enable. Set KEYWEN=1 after key stable, and then enter sleep mode.	0 = key wakeup disable and clear flag 1 = key wakeup enable;

9.4 0x0031, P_WAKEUP_Flag

P_WAKEUP_Flag			0x0031			Wake up Control Flag Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	TBHWFC	-	TBLWFC	KEYWFC
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
3	TBHWFC	R	32K Time Base High Frequency Wakeup source Flag Disable TBHEN to clear TBHWFC	Read 0 = TBH wakeup not happen Read 1 = TBH wakeup occurred
2	-	-	Reserved	-
1	TBLWFC	R	32K Time Base Low Frequency Wakeup source Flag Disable TBLWFC to clear TBLWFC	Read 0 = TBL wakeup not happen Read 1 = TBL wakeup occurred
0	KEYWFC	R	Key Wakeup source Flag	Read 0 = key wakeup not happen

Bit	Function	Type	Description	Condition
			Disable KEYWEN to clear KEYWFC	Read 1 = key wakeup occurred

9.5 0x0037, P_IO_PortB_WakeUpEN

P_IO_PortB_WakeUpEN			0x0037		PortB Wakeup Control Register			
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	PBWK3	PBWK2	PBWK1	PBWK0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	-	-	Reserved	-
[3:0]	PBWK[3:0]	R/W	PortB[3:0] key change wakeup enable	0: disable wakeup function 1: enable wakeup function

9.6 0x0038, P_IO_PortC_WakeUpEN

P_IO_PortC_WakeUpEN			0x0038		PortC Wakeup Control Register			
Bit	7	6	5	4	3	2	1	0
Function	PCWK7	PCWK6	PCWK5	PCWK4	-	-	PCWK1	PCWK0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	PCWK[7:4]	R/W	PortC[7:4] key change wakeup enable.	0: disable wakeup function 1: enable wakeup function
[3:2]	-	-	Reserved	-
[1:0]	PCWK[1:0]	R/W	PortC[1:0] key change wakeup enable.	0: disable wakeup function 1: enable wakeup function

9.7 0x0039, P_IO_PortD_WakeUpEN

P_IO_PortD_WakeUpEN			0x0039		PortD Wakeup Control Register			
Bit	7	6	5	4	3	2	1	0
Function	PDWK7	-	PDWK5	-	PDWK3	PDWK2	-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	PDWK7	R/W	PortD[7] key change wakeup enable.	0: disable wakeup function 1: enable wakeup function
6	-	-	Reserved	-
5	PDWK5	R/W	PortD[5] key change wakeup enable.	0: disable wakeup function 1: enable wakeup function
4	-	-	Reserved	-
[3:2]	PDWK[3:2]	R/W	PortD[3:2] key change wakeup enable.	0: disable wakeup function

Bit	Function	Type	Description	Condition
				1: enable wakeup function
[1:0]	-	-	Reserved	-

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10 RFC FUNCTION

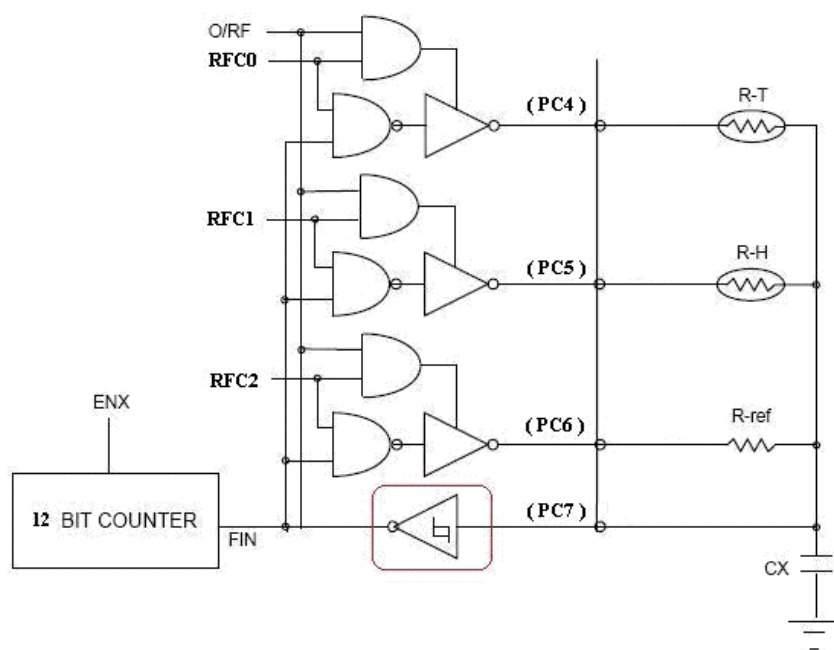
10.1 Introduction

The GPL812P01A features a Resistor to Frequency Converter (RFC) function. Because the heat-resistance sensor and moisture-resistance sensor will change its resistance due to the change of temperature and humidity, RC time constant will also be changed. Programmers can use this identity to measure temperature or humidity.

10.2 0x001E, P_RFC_Ctrl

P_RFC_Ctrl			0x001E			RFC Control Register		
Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	RFCEN	RFC2	RFC1	RFC0
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[7:4]	Reserved	Reserved	-	-
3	RFCEN	R/W	This bit can set RFC function as enable or disable.	0: RFC function disable 1: RFC function enable
[2:0]	RFC[2:0]	R/W	RFC[2:0] are RFC enable function with PortC[6:4].	RFC2 = 1, PC6 RFC2 enable RFC2 = 0, PC6 RFC2 disable RFC1 = 1, PC5 RFC1 enable RFC1 = 0, PC5 RFC1 disable RFC0 = 1, PC4 RFC0 enable RFC0 = 0, PC4 RFC0 disable



In RFC application, PortC[6:4] can connect to sensor and capacitor.

10.3 How to Use RFC Function

- Step 1: Set PortC as input floating status.
- Step 2: Enable RFC function
- Step 3: Enable Timer1 and set Timer1 clock source from PC.7
- Step 4: Sequence enable RFC2/RFC1/RFC0
- Step 5: Record Timer1 count data
- Step 6: Repeat Step4

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11 VOLTAGE DETECTOR

11.1 Introduction

The voltage detector comprises a band-gap reference generator and a comparator. The band-gap reference generates a fixed voltage ($\approx 2.1V/2.4V/2.7V$) for comparator to sense a low voltage event. There are three programmable detection levels provided. Program can be written to monitor the Register 0x0033 for low voltage event detection.

The detector checks VDD real time to sense the low voltage event. If the power supply voltage drops below the detection level, the LVD flag will be generated to inform CPU the low voltage event occurs. The Low Voltage Reset (LVR) of GPL812P01A is always enabled. When VDD drops below LVR level for more than 2~3 μ s, LVR will reset the device.

11.2 0x0033, P_LVD_Ctrl

P_LVD_Ctrl			0x0033		Low Voltage Detector Control			
Bit	7	6	5	4	3	2	1	0
Function	LV DEN	LV DS[1]	LV DS[0]	-	-	-	-	LVD
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
7	LV DEN	R/W	Enable / Disable voltage detector	0= disable 1= enable
[6:5]	LV DS[1:0]	R/W	Low voltage level selection	00 = 2.1V, LVD=1, VDD<2.1V trigger LVD 01 = 2.4V, LVD=1, VDD<2.4V trigger LVD 10 or 11 = 2.7V LVD=1, VDD<2.7V trigger LVD. (2.7V only for GPL812P01A, EV chip not support)
[4:1]	-	-	Reserved	-
0	LVD	R	Low voltage detect flag	Read 0= low voltage not detected Read 1= low voltage detected

12 HARDWARE SUMMARY

12.1 GPL812P01A Mapping

0x0000-0x007F	Control registers
0x0080-0x00FF	SRAM
0x0100-0x01BF	Reserved
0x01C0-0x01FF	SRAM
0x0200-0x3FFF	Reserved
0x6000-0x7FFF	ROM (bank)L
0x8000-0xDFFF	Reserved
0xE000-0xFFFF	Always mapping into 0x6000-0x7FFF

12.2 GPL812P01A Control Register

Address	Function	Reset value	R/W	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x0000	P_IO_PortB_Dir	00h	R/W	-	-	-	-	IOBDir			
0x0001	P_IO_PortC_Dir	00h	R/W	IOCDIR				-	-	IOCDIR	
0x0002	P_IO_PortD_Dir	00h	R/W	IODDIR	-	IODDIR	-	IODDIR		-	-
0x0003	P_IO_PortA_Attrib	00h	R/W	IOAAttrib							
0x0004	P_IO_PortB_Attrib	00h	R/W	-	-	-	-	IOBAttrib			
0x0005	P_IO_PortC_Attrib	00h	R/W	IOCAattrib				-	-	IOCAattrib	
0x0006	P_IO_PortD_Attrib	00h	R/W	IODAttrib	-	IODAttrib	-	IODAttrib		-	-
0x0007	P_IO_PortA_Data	00h	R/W	IOADData							
0x0008	P_IO_PortB_Data	00h	R/W	-	-	-	-	IOBData			
0x0009	P_IO_PortC_Data	00h	R/W	IOCData				-	-	IOCData	
0x000A	P_IO_PortD_Data	00h	R/W	IODData	-	IODData	-	IODData		-	-
0x000B	P_IO_PortA_Dir	00h	R/W	IOADIR							
0x000C	P_IO_PortA_Buffer	00h	R	IOABUF							
0x000D	P_IO_PortB_Buffer	00h	R	-	-	-	-	IOBBUF			
0x000E	P_IO_PortC_Buffer	00h	R	IOCBUF				-	-	IOCBUF	
0x000F	P_IO_PortD_Buffer	00h	R	IODBUF	-	IODBUF	-	IODBUF		-	-
0x0011	P_IO_Green_Ctrl	00h	W	-	-	Green mode	-	-	-	-	-
0x0012	P_SYSTEM_Ctrl	00h	W	SLEEP = AAH , Write other data system reset							
0x0017	P_SYSTEM_Green_Ctrl	00h	W	-	Green mode	-	-	-	-	-	-
0x0013	P_INT_Ctrl0	00h	R/W	0	TM0O_IN T	0	TM1O_INT	CPUDiv1K_INT	CPUDiv 4K_INT	CPUDiv3 2K_INT	CPUDiv2M_I NT
0x0014	P_INT_Clear0	00h	R/W	-	TM0O_Cl r	-	TM1O_Clr	CPUDiv1K_Clr	CPUDiv 4K_Clr	CPUDiv3 2K_Clr	CPUDiv2M_ Clr
0x0015	P_INT_Ctrl1	00h	R/W	-	-	TBH_INT	TBL_INT	-	-	-	-

Address	Function	Reset value	R/W	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
				Bit [5] 32K Time Base high frequency interrupt 0 = disable(Default) 1 = enable Bit [4] 32K Time Base low frequency interrupt 0 = disable(Default) 1 = enable							
0x0016	P_INT_Clear1	00h	R/W	-	-	TBH_Clr	TBL_Clr	-	-	-	-
				Bit [5] 32K Time Base high frequency interrupt Flag Write 1 to clear Bit [4] 32K Time Base low frequency interrupt Flag Write 1 to clear							
0x0018	P_INT_Ctrl2	00h	R/W	0	-	0	-	-	-	EXT_INT1	EXT_INT0
				Bit[0,1] key change will trigger interrupt.							
0x0019	P_INT_Clear2	00h	R/W	-		-	-	-	-	PB1_Clr	PB0_Clr
				Bit[0,1] 1 = interrupted 0 = non interrupt							
0x001E	P_RFC_Ctrl	00h	R/W	-	-	-	-	RFCEN	RFC2	RFC1	RFC0
								RFC function 0 = disable 1 = enable	RFC2/1/0 is mapping to PC6/PC5/PC4		
0x0020	P_WDT_Flag_Clear	00h	W	WDTCLEAR = AAH							
0x0021	P_TIMER_Timer0_Ctrl	00h	R/W	TM0ES	-	TM0CLK1	TM0CLK0	-	-	-	-
				Bit [7] TM0ES: Timer0 enable/disable control. 0 = disable; 1 = enable. Bit [5:4] TM0CLK[1:0]: Timer0 clock source select bits 00 = Fosc 01 = Fosc/2 10 = Fosc/4 11 = Fosc/16							
0x0022	P_TIMER_Timer1_Ctrl	00h	R/W	TM1ES	TM1CLK1 O	TM1CLK1	TM1CLK0	-	-	-	0

Address	Function	Reset value	R/W	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
				Bit [7] TM1ES: Timer1 enable/disable control. 0 = disable; 1 = enable. Bit [6] Timer1 clock source from IO selection. 0 = Timer1 clock source setting from bit4/5. 1 = Timer1 clock source from IO portC[7]. (RFC using) Bit [5:4] Timer1 clock source setting from bit4/5. 00 = Fosc 01 = Fosc/4 10 = Fosc/64 11 = TM0OF							
0x0023	P_TIMER_TM0Data_LB	FFh	R	8bits timer : TM0 Counter 8 Bits							
			W	8bits timer : TM0 8 Bits Register							
0x0025	P_TIMER_TM1Data_LB	00h	R	TM1 Counter Low 8 Bits							
			W	TM1 Low 8 Bits Register							
0x0026	P_TIMER_TM1Data_HB	00h	R	-	-	-	-	TM1 Counter High 4 Bits			
			W					TM1 High 4 Bits Register			
0x002A	P_CLK_32768_En	00h	R/W	EN	-	0	-	-	0	0	0
0x002B	P_TIMER_TimeBase_Sel	00h	R/W	-	-	-	32kStr_T	TBLSEL1	TBLSEL0	TBHSEL1	TBHSEL0
0x002C	P_IO_Strobe_En	00h	R/W	-	-	-	-	-	-	-	StrobeEn
0x002D	P_IO_PortB_Strobe_En	00h	R/W	-	-	-	-	STRB3	STRB2	STRB1	STRB0
0x002E	P_IO_PortC_Strobe_En	00h	R/W	STRC7	STRC6	STRC5	STRC4	-	-	STRC1	STRC0
0x002F	P_IO_PortD_Strobe_En	00h	R/W	STRD7	-	STRD5	-	STRD3	STRD2	-	-
0x0030	P_WAKEUP_Ctrl	00h	R/W	-	-	-	-	TBHWEN	-	TBLWEN	KEYWEN
0x0031	P_WAKEUP_Flag	00h	R	-	-	-	-	TBHWFC	-	TBLWFC	KEYWFC
0x0032	P_LCD_Ctrl	00h	R/W	Green Mode1	VLCDSEL1	VLCDSEL0	Green Mode	LBSDT1	LBSDT0	LCDMD1	LCDMD0
0x0033	P_LVD_Ctrl	00h	R/W	LV DEN	LVDS1	LVDS0	-	-	-	-	LVD
0x0037	P_IO_PortB_WakeUpEN	00h	R/W	-	-	-	-	PBWK3	PBWK2	PBWK1	PBWK0
0x0038	P_IO_PortC_WakeUpEN	00h	R/W	PCWK7	PCWK6	PCWK5	PCWK4	-	-	PCWK1	PCWK0
0x0039	P_IO_PortD_WakeUpEN	00h	R/W	PDWK7	-	PDWK5	-	PDWK3	PDWK2	-	-
0x003F	P_SRAM_Ctrl	00h	R/W	-	-	page 5	-	-	-	page 1	page 0
0x004A	P_IOA_LCDPORT_Ctrl	00h	R/W	IOA_LCD En_7	IOA_LCD En_6	IOA_LCD En_5	IOA_LCD En_4	IOA_LCD En_3	IOA_LCD En_2	IOA_LCD En_1	IOA_LCD En_0
0x0048	P_IOA_SEGCOM_Ctrl	00h	R/W	IOA_SEG COM_7	IOA_SEG COM_6	-	-	-	-	-	-
0x004B	P_IOB_LCDPORT_Ctrl	00h	R/W	-	-	-	-	IOB_LCD En_3	IOB_LCD En_2	IOB_LCD En_1	IOB_LCD En_0

Address	Function	Reset value	R/W	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x0049	P_IOB_SEGCOM_Ctrl	00h	R/W	-	-	-	-	IOC_SEGC OM_3	IOC_SE GCOM_ 2	IOC_SEG COM_1	IOC_SEGCO M_0
0x004C	P_IOC_LCDPORT_Ctrl	00h	R/W	IOC_LCD En_7	IOC_LCD En_6	IOC_LCD En_5	IOC_LCDEn _4	~	~	IOC_LCD En_1	IOC_LCDEn_ 0
0x0034	P_IOC_SEGCOM_Ctrl	00h	R/W	IOC_SEG COM_7	IOC_SEG COM_6	IOC_SEG COM_5	IOC_SEGC OM_4	-	-	IOC_SEG COM_1	IOC_SEGCO M_0
0x0035	P_IOD_LCDPORT_Ctrl	00h	R/W	-	-	-	-	IOD_SEGE n_3	IOD_SE GEn_2	-	-

Note: If the bits of the register are not defined, they will not be implemented in the real chip and the read value should be random. However, it is defined as 0 in this table for simplification.

13 APPENDIX A. INSTRUCTION SET

No.	Syntax	Code No.	Code Size	CPU Cycle	Addressing Mode
001	ADC #dd	69H	2	2	immediate
002	ADC aa	65H	2	3	zero page
003	ADC aa,X	75H	2	4	zero page indexed x
004	ADC aaaa	6DH	3	4	absolute
005	ADC aaaa,X	7DH	3	4	absolute indexed x
006	ADC aaaa,Y	79H	3	4	absolute indexed y
007	ADC (aa,X)	61H	2	6	indexed indirect x
008	ADC (aa),Y	71H	2	5	indirect indexed y
009	AND #dd	29H	2	2	immediate
010	AND aa	25H	2	3	zero page
011	AND aa,X	35H	2	4	zero page indexed x
012	AND aaaa	2DH	3	4	absolute
013	AND aaaa,X	3DH	3	4	absolute indexed x
014	AND aaaa,Y	39H	3	4	absolute indexed y
015	AND (aa,X)	21H	2	6	indexed indirect x
016	AND (aa),Y	31H	2	5	indirect indexed y
017	ASL A	0AH	1	2	accumulator
018	ASL aa	06H	2	5	zero page
019	ASL aa,X	16H	2	6	zero page indexed x
020	ASL aaaa	0EH	3	6	absolute
021	ASL aaaa,X	1EH	3	7	absolute indexed x
022	BCC ??	90H	2	2/3/4*	relative
023	BCS ??	B0H	2	2/3/4*	relative
024	BEQ ??	F0H	2	2/3/4*	relative
025	BIT aa	24H	2	3	zero page
026	BIT aaaa	2CH	3	4	absolute
027	BMI ??	30H	2	2/3/4*	relative
028	BNE ??	D0H	2	2/3/4*	relative
029	BPL ??	10H	2	2/3/4*	relative
030	BRK	00H	1	7	implied
031	BVC ??	50H	2	2/3/4*	relative
032	BVS ??	70H	2	2/3/4*	relative
033	CLC	18H	1	2	implied
034	CLD	D8H	1	2	implied
035	CLI	58H	1	2	implied
036	CLV	B8H	1	2	implied
037	CMP #dd	C9H	2	2	immediate
038	CMP aa	C5H	2	3	zero page
039	CMP aa,X	D5H	2	4	zero page indexed x
040	CMP aaaa	CDH	3	4	absolute
041	CMP aaaa,X	DDH	3	4	absolute indexed x
042	CMP aaaa,Y	D9H	3	4	absolute indexed y
043	CMP (aa,X)	C1H	2	6	indexed indirect x
044	CMP (aa),Y	D1H	2	5	indirect indexed y
045	CPX #dd	E0H	2	2	immediate
046	CPX aa	E4H	2	3	zero page
047	CPX aaaa	ECH	3	4	absolute
048	CPY #dd	C0H	2	2	immediate
049	CPY aa	C4H	2	3	zero page
050	CPY aaaa	CCH	3	4	absolute
051	DEC aa	C6H	2	5	zero page
052	DEC aa,X	D6H	2	6	zero page indexed x
053	DEC aaaa	CEH	3	6	absolute
054	DEC aaaa,X	DEH	3	7	absolute indexed x
055	DEX	CAH	1	2	implied
056	DEY	88H	1	2	implied
057	EOR #dd	49H	2	2	immediate
058	EOR aa	45H	2	2	zero page
059	EOR aa,X	55H	2	2	zero page indexed x
060	EOR aaaa	4DH	3	3	absolute

No.	Syntax	Code No.	Code Size	CPU Cycle	Addressing Mode
061	EOR aaaa,X	5DH	3	3	absolute indexed x
062	EOR aaaa,Y	59H	3	3	absolute indexed y
063	EOR (aa,X)	41H	2	2	indexed indirect x
064	EOR (aa),Y	51H	2	2	indirect indexed y
065	INC aa	E6H	2	5	zero page
066	INC aa,X	F6H	2	6	zero page indexed x
067	INC aaaa	EEH	3	6	absolute
068	INC aaaa,X	FEH	3	7	absolute indexed x
069	INX	E8H	1	2	implied
070	INY	C8H	1	2	implied
071	JMP aaaa	4CH	3	3	absolute
072	JMP (aaaa)	6CH	3	5	indirect absolute
073	JSR aaaa	20H	3	6	absolute
074	LDA #dd	A9H	2	2	immediate
075	LDA aa	A5H	2	3	zero page
076	LDA aa,X	B5H	2	4	zero page indexed x
077	LDA aaaa	ADH	3	4	absolute
078	LDA aaaa,X	BDH	3	4	absolute indexed x
079	LDA aaaa,Y	B9H	3	4	absolute indexed y
080	LDA (aa,X)	A1H	2	6	indexed indirect x
081	LDA (aa),Y	B1H	2	5	indirect indexed y
082	LDX #dd	A2H	2	2	immediate
083	LDX aa	A6H	2	3	zero page
084	LDX aa,Y	B6H	2	4	zero page indexed y
085	LDX aaaa	AEH	3	4	absolute
086	LDX aaaa,Y	BEH	3	4	absolute indexed y
087	LDY #dd	A0H	2	2	immediate
088	LDY aa	A4H	2	3	zero page
089	LDY aa,X	B4H	2	4	zero page indexed x
090	LDY aaaa	ACH	3	4	absolute
091	LDY aaaa,X	BCH	3	4	absolute indexed x
092	LSR A	4AH	1	2	accumulator
093	LSR aa	46H	2	5	zero page
094	LSR aa,X	56H	2	6	zero page indexed x
095	LSR aaaa	4EH	3	6	absolute
096	LSR aaaa,X	5EH	3	7	absolute indexed x
097	NOP	EAH	1	2	implied
098	ORA #dd	09H	2	2	immediate
099	ORA aa	05H	2	3	zero page
100	ORA aa,X	15H	2	4	zero page indexed x
101	ORA aaaa	0DH	3	4	absolute
102	ORA aaaa,X	1DH	3	4	absolute indexed x
103	ORA aaaa,Y	19H	3	4	absolute indexed y
104	ORA (aa,X)	01H	2	6	indexed indirect x
105	ORA (aa),Y	11H	2	5	indirect indexed y
106	PHA	48H	1	3	implied
107	PHP	08H	1	3	implied
108	PLA	68H	1	4	implied
109	PLP	28H	1	4	implied
110	ROL A	2AH	1	2	accumulator
111	ROL aa	26H	2	5	zero page
112	ROL aa,X	36H	2	6	zero page indexed x
113	ROL aaaa	2EH	3	6	absolute
114	ROL aaaa,X	3EH	3	7	absolute indexed x
115	ROR A	6AH	1	2	accumulator
116	ROR aa	66H	2	5	zero page
117	ROR aa,X	76H	2	6	zero page indexed x
118	ROR aaaa	6EH	3	6	absolute
119	ROR aaaa,X	7EH	3	7	absolute indexed x
120	RTI	40H	1	6	implied
121	RTS	60H	1	6	implied
122	SBC #dd	E9H	2	2	immediate
123	SBC aa	E5H	2	3	zero page
124	SBC aa,X	F5H	2	4	zero page indexed x
125	SBC aaaa	EDH	3	4	absolute

No.	Syntax	Code No.	Code Size	CPU Cycle	Addressing Mode
126	SBC aaaa,X	FDH	3	4	absolute indexed x
127	SBC aaaa,Y	F9H	3	4	absolute indexed y
128	SBC (aa,X)	E1H	2	6	indexed indirect x
129	SBC (aa),Y	F1H	2	5	indirect indexed y
130	SEC	38H	1	2	implied
131	SED	F8H	1	2	implied
132	SEI	78H	1	2	implied
133	STA aa	85H	2	3	zero page
134	STA aa,X	95H	2	4	zero page indexed x
135	STA aaaa	8DH	3	4	absolute
136	STA aaaa,X	9DH	3	5	absolute indexed x
137	STA aaaa,Y	99H	3	5	absolute indexed y
138	STA (aa,X)	81H	2	6	indexed indirect x
139	STA (aa),Y	91H	2	6	indirect indexed y
140	STX aa	86H	2	3	zero page
141	STX aa,Y	96H	2	4	zero page indexed y
142	STX aaaa	8EH	3	4	absolute
143	STY aa	84H	2	3	zero page
144	STY aa,X	94H	2	4	zero page indexed x
145	STY aaaa	8CH	3	4	absolute
146	TAX	AAH	1	2	implied
147	TAY	A8H	1	2	implied
148	TSX	BAH	1	2	implied
149	TXA	8AH	1	2	implied
150	TXS	9AH	1	2	implied
151	TYA	98H	1	2	implied

* If there is no branch jump, the CPU cycle is 2. If branch jump condition is confirmed and jumps to the same page, the CPU cycle is 3. If it jumps to different page, the CPU cycle is 4.